

# TEXAS PACIFIC LAND TRUST TPL

May 09, 2019 - 1:28pm EST

by

Plainview

|                      |        |          |       |       |
|----------------------|--------|----------|-------|-------|
|                      |        |          | 2019  | 2020  |
| Price:               | 780.00 | EPS      | 31.92 | 39.31 |
| Shares Out. (in M):  | 8      | P/E      | 24    | 20    |
| Market Cap (in \$M): | 6,050  | P/FCF    | 29    | 24    |
| Net Debt (in \$M):   | -93    | EBIT     | 305   | 375   |
| TEV (in \$M):        | 5,957  | TEV/EBIT | 20    | 16    |

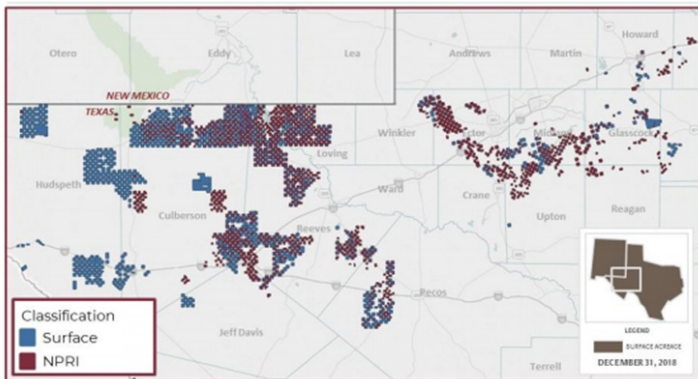
## Description

TPL has previously been written up and discussed on VIC, but I believe that this analysis adds to the discussion and is worth a full writeup. When I started this exercise, I thought this was going to be a short sale recommendation; TPL is an oil and gas company that trades at ~21x Q119 annualized EBITDA. However, after doing the work, I believe that TPL is a long.

TPL is a liquidating trust that was set up in 1888 to manage the assets of the bankrupt Texas & Pacific Railway Company. TPL has the ability to manage its assets with all the powers of an absolute owner, but it is not authorized to issue new equity. The business model of the Trust, in accordance with the trust indenture, is to manage the land and mineral rights and to divest those lands and mineral rights on an opportunistic basis, eventually liquidating all assets for the benefit of investors. Over the past few years, TPL has gotten more active, forming a water company and reinvesting the proceeds of asset sales instead of distributing all of the proceeds to investors. I will talk a little about governance and the current proxy fight later on. It is a cause for concern. Trustees serve for life, severely diminishing the ability for shareholders to exert influence.

TPL has three main buckets of value: surface acreage/rights, water business, perpetual royalty interests. Below is a decent map of TPL's acreage position.

### Royalty and Surface Acreage Position



### Surface Acreage/Rights

The simplest bucket of value to deal with is the surface acreage/rights. TPL's surface rights generate revenue from easement contracts covering activities such as oil and gas pipelines and subsurface wellbore easements. The majority of TPL's easements have a ten-year term (i.e. TPL gets paid again every ten years if the easement is still required). TPL also leases land to operators and midstream companies for facilities and roads. From time to time TPL will sell pieces of their surface acreage. Typically water rights are attached to the surface rights which makes it hard to separate the value of the two when looking at recent land transactions; however an attempt to do so must be made to avoid double counting the value of the surface and the value of the water business. Below is a table of recent TPL land sales and acquisitions and the implied \$/acre metrics. There is a wide range. Anecdotally I have heard that pure goat pasture in the middle of nowhere West TX is ~\$400/acre. The lowest value in the table below for the Hudspeth and Culberson sale appears to confirm that figure as being conservative. However, there are definitely parcels of land closer to oil and gas activity or small metropolitan areas where surface rights ex water would have more value than that.

| Land Sales/Acquisitions       | Acres  | \$mm   | \$/acre   |
|-------------------------------|--------|--------|-----------|
| Q219 - Loving County          | 3,400  | \$ 20  | \$ 6,000  |
| Q119 - Reeves and Loving      | 14,002 | \$ 100 | \$ 7,142  |
| Q119 - Hudspeth and Culberson | 7,249  | \$ 4   | \$ 497    |
| 2018 Land Sales               | 172    | \$ 4   | \$ 25,656 |
| 2018 Land Acquisitions        | 14,650 | \$ 9   | \$ 642    |

Below you can see my assumptions in coming up with a blended average surface rights value range of \$560/acre to \$800/acre.

| Surface Value \$/acre Estimates (excludes water rights) |             |     |              |         |
|---------------------------------------------------------|-------------|-----|--------------|---------|
|                                                         | Low \$/acre |     | High \$/acre |         |
| Land Value                                              | \$          | 400 | \$           | 2,000   |
|                                                         |             |     |              | Blended |
| Low                                                     |             | 90% | 10%          | \$ 560  |
| High                                                    |             | 75% | 25%          | \$ 800  |

Using those blended averages, I calculated the pre-tax surface rights value to be between ~\$500mm and ~\$700mm (see table below), which implies a ~10x to ~14x multiple on the past 3 years average easement revenue ex water easements. Assuming no tax basis, that is ~\$400mm to \$565mm of after tax liquidation value to TPL.

| Surface Acreage Value (excluding water rights) |     |         |      |         |
|------------------------------------------------|-----|---------|------|---------|
|                                                | Low |         | High |         |
| Assumed \$/Acre                                | \$  | 560     | \$   | 800     |
| Surface Acres                                  |     | 896,027 |      | 896,027 |
| Surface Acreage Value                          | \$  | 502     | \$   | 717     |
| Implied Mult. of 3 Yr Avg Ease Rev             |     | 9.8x    |      | 13.9x   |
| After Tax Surface Acreage Value                | \$  | 396     | \$   | 566     |

It is worth noting that TPL is starting to get more active in this business line. They sold a large parcel at an apparently nice valuation to WPX in January 2019 for \$100mm. Rather than pay the tax and distribute the proceeds to shareholders, management decided to structure the transaction as a 1031 exchange to defer the taxes and reinvest the proceeds. This is not necessarily bad, but it is out of line with the stated purpose of the trust and bears watching, especially given the aforementioned governance concerns.

### Water Business

Over the past couple of years, TPL has started building out a team to take advantage of the water rights associated with its ~900k surface acres. The water opportunity in the Permian, and especially the Delaware, is huge. There are multiple private equity teams with massive equity commitments chasing deals (Goodnight, Waterbridge, etc). TPL's water rights are extremely valuable and provide TPL with a competitive advantage in the water sourcing and disposal businesses. Most industry participants must make deals with surface owners for water sourcing rights, water disposal rights, and water pipeline easements. Typically these agreements involve up front payments and a royalty payment per barrel of water. TPL's vast surface ownership gives it a cost advantage over its competitors when it chooses to compete head to head for customers because TPL does not have to pay anything (up front or royalty) to the surface owner. TPL is the surface owner. In addition, when the economics make sense, TPL can choose to partner with a competitor who will spend all required capital and pay TPL a royalty on any water produced from or disposed on TPL's land. In short, TPL's water business is far more capital efficient than its competitors. Since forming Texas Pacific Water Resources LLC in June 2017, TPL has invested ~\$70mm in the water business. Q119 annualized EBITDA was \$39mm (100% G&A burden) which implies a ~2.0x EBITDA multiple. Those are strong economics. TPL is hiring aggressively to build out its water team (employee count increased from 8 at 12/31/16 to 71 at 1/31/19). Getting the right commercial team in place is a key risk to this business, but TPL is making good progress. Robert Crain joined TPL in June 2017 as the Executive Vice President of TPWR. Prior to joining TPL, Crain was Water Resources Manager with EOG Resources where he led the development of EOG's water resource programs in the Eagle Ford and Permian.

Below I have set a valuation range for the Water Business at 20.0x - 30.0x Q119 annualized EBITDA, and once again I have applied a 21% tax rate assuming no tax basis to get to a net liquidation value to TPL.

| Water Business Value      |     |       |      |       |
|---------------------------|-----|-------|------|-------|
|                           | Low |       | High |       |
| Assumed EBITDA Multiple   |     | 20.0x |      | 30.0x |
| Q119 Ann. EBITDA          | \$  | 39    | \$   | 39    |
| Water Business Value      | \$  | 784   | \$   | 1,176 |
| After Tax Water Biz Value | \$  | 619   | \$   | 929   |

At first glance these EBITDA multiples appear to be insane. The only publicly traded source water business (WTTR) trades at a mid single digit EBITDA multiple and the produced water businesses changing hands in the private market seem to be in the high single digit to low double digit EBITDA multiple range. That said, I believe that 20.0x - 30.0x is correct. In the tables below I have laid out two simple DCF's for the water business. In the top table, the High Case, TPL invests 100% of its Water Business EBITDA each year back into the business at a 2.5x multiple through 2023 at which point I applied a 9.0x multiple. In the low case we make the same assumptions except we are investing capital at a 4.0x multiple. The PV8s of these two simple DCFs line up pretty closely to the pre-tax values implied by the EBITDA multiple range assumed above.

| Water Business DCF - High Case |          |       |       |        |           |
|--------------------------------|----------|-------|-------|--------|-----------|
| Capital Investment Multiple    |          | 2.5x  |       |        |           |
|                                | 2019     | 2020  | 2021  | 2022   | 2023 / TY |
| EBITDA                         | \$ 50    | \$ 70 | \$ 98 | \$ 137 | \$ 192    |
| Capital                        | (50)     | (70)  | (98)  | (137)  | 9.0x      |
| FCF                            | -        | -     | -     | -      | \$ 1,729  |
| PV8 - High Case                | \$ 1,177 |       |       |        |           |

| Water Business DCF - Low Case |        |       |       |       |           |
|-------------------------------|--------|-------|-------|-------|-----------|
| Capital Investment Multiple   |        | 4.0x  |       |       |           |
|                               | 2019   | 2020  | 2021  | 2022  | 2023 / TY |
| EBITDA                        | \$ 50  | \$ 63 | \$ 78 | \$ 98 | \$ 137    |
| Capital                       | (50)   | (63)  | (78)  | (98)  | 9.0x      |
| FCF                           | -      | -     | -     | -     | \$ 1,230  |
| PV8 - Low Case                | \$ 837 |       |       |       |           |

But how do we know that TPL will be able to invest that much capital efficiently? Maybe those terminal year EBITDAs and investment multiples are unreasonable. The table below lays out one example of how TPL's water business could generate those EBITDA figures and the market share implications. Note that TPL does not provide any operating stats for its water business so we must make educated assumptions about volumes and unit economics. An assumed blended margin per bbl of water of 35c/bbl (higher for operated assets, lower for royalty bbls) implies 1.1mmb/d to 1.5mmb/d of water volume net to TPL. For illustration, let's split that volume 40/60 between the source water business (which I assume includes their recycling business) and the produced water business. We now have 428kb/d to 601kb/d of source water and 642kb/d to 902kb/d of produced water. Let's take the 428kb/d to 601kb/d of source water first. Assuming 400kbb/d of water required per completion (per IHS and industry sources, that is the Delaware basin avg), that equates to 391 to 549 completions / year. At 12 completions per rig per year, that is 33 to 46 rigs. The Delaware rig count is ~250. That is a 13% to 18% source water market share which seems very reasonable when you look at a map of TPL's water rights and historical produced water business EBITDA growth rate. The future 2023 market share will probably be even lower given that the Delaware rig count will almost certainly grow over that time period (increase production = increased cash flow = more rigs). With respect to the produced water side, the figures below imply a 8% to 12% market share of IHS's 2023 forecasted Delaware basin produced water volume excluding EOR volumes.

|                              | Low     | High    |
|------------------------------|---------|---------|
| Terminal Year EBITDA         | \$ 137  | \$ 192  |
| Estimated Margin/bbl         | \$ 0.35 | \$ 0.35 |
| Required Water Volume (kb/d) | 1,070   | 1,504   |
| Source Water                 | 40% 428 | 601     |
| Produced Water               | 60% 642 | 902     |

#### Completion Market Share

|                                  |         |         |
|----------------------------------|---------|---------|
| TPL Source Water Required (kb/d) | 428     | 601     |
| Water per Completion (bbls)      | 400,000 | 400,000 |
| Completions/Year                 | 391     | 549     |
| Wells per Rig / Year             | 12      | 12      |
| Required Rigs                    | 33      | 46      |
| Delaware Basin Rig Count         | 250     | 250     |
| Market Share                     | 13%     | 18%     |

#### Produced Water Market Share

|                                               |       |       |
|-----------------------------------------------|-------|-------|
| TPL Produced Water Required (kb/d)            | 642   | 902   |
| 2023 Delaware Produced Water per IHS excl EOR | 7,726 | 7,726 |
| 2023 Market Share                             | 8%    | 12%   |

One final note on the water business is recycling. Recycling is detrimental to the value of water rights, and we need to continue to monitor its adoption; however, I believe that significant fresh source water will continue to be required absent significant regulation against it (which in TX is not very likely). The source water business is driven by logistics. You need the right amount of water in the right place (right flow rate) at the right time. Having source water wells dotted across the Delaware sweet spot is going to be valuable in almost all future scenarios. Furthermore, the customer relationships built by TPL's fresh source water business and produced water business will be an important advantage as they grow their exposure to the recycled water business.

#### Non Participating Royalty Interests

TPL owns a 1/128 royalty in ~84,934 acres and a 1/16 royalty in 370,737 acres. To get a more meaningful number I multiply the acres by the royalty fraction to get net royalty acres of 23,835 (100% NRI). In addition, TPL acquired ~228 NRA's in 2018 and 38 NRA's YTD in 2019 for a total of 24,100 NRAs at 3/31/19. Some of this acreage is in the core of the Permian Basin (mostly Delaware Basin). But not everything is prospective for oil and gas. Below I lay out my prospectivity assumptions that underpin the well inventory and EUR analysis discussed later. There is nothing magic here. I looked at the map and made a judgement.

## TPL Nonparticipating Royalty Interests

|                  | 0.78%  | 6.25%   | NRA    | Assumed<br>Prospectivity |
|------------------|--------|---------|--------|--------------------------|
| 2018 Acquisition |        |         | 228    | 95%                      |
| 2019 Acquisition |        |         | 38     | 95%                      |
| Callahan         | -      | 80      | 5      | 0%                       |
| Coke             | -      | 1,184   | 74     | 0%                       |
| Concho           | -      | -       | -      | 0%                       |
| Crane            | 265    | 5,198   | 327    | 10%                      |
| Culberson        | -      | 111,513 | 6,970  | 60%                      |
| Ector            | 33,633 | 11,793  | 1,000  | 25%                      |
| El Paso          | -      | -       | -      | 0%                       |
| Fisher           | -      | 320     | 20     | 0%                       |
| Glasscock        | 3,600  | 11,111  | 723    | 50%                      |
| Howard           | 3,099  | 1,840   | 139    | 80%                      |
| Hudspeth         | -      | 1,008   | 63     | 5%                       |
| Jeff Davis       | -      | 7,555   | 472    | 0%                       |
| Loving           | 6,107  | 48,066  | 3,052  | 95%                      |
| Midland          | 12,945 | 13,120  | 921    | 95%                      |
| Mitchell         | 1,760  | 586     | 50     | 10%                      |
| Nolan            | 2,488  | 3,157   | 217    | 0%                       |
| Palo Pinto       | -      | 800     | 50     | 0%                       |
| Pecos            | 320    | 16,895  | 1,058  | 20%                      |
| Presidio         | -      | 3,200   | 200    | 0%                       |
| Reagan           | 6,162  | 1,274   | 128    | 0%                       |
| Reeves           | 3,013  | 116,691 | 7,317  | 90%                      |
| Stephens         | 2,817  | 160     | 32     | 0%                       |
| Sterling         | 640    | 2,080   | 135    | 0%                       |
| Taylor           | -      | 966     | 60     | 0%                       |
| Upton            | 6,903  | 9,101   | 623    | 95%                      |
| Winkler          | 1,182  | 3,040   | 199    | 10%                      |
| Total            | 84,934 | 370,737 | 24,100 | 68.0%                    |

It is worth noting that Chevron and Anadarko are the two largest operators underpinning the Trust's royalty production accounting for 22% and 18%, respectively, of TPL's royalty revenue in 2018. CVX recently laid out their ambitious long term plans for the Permian, and TPL shareholders will get to go along for the ride free of cost. I view the OXY/APC merger as neutral to negative to TPL. OXY is a very strong Permian operator and will have the pro forma scale required to underpin an efficient multi-well pad development program; however, the pro forma debt / preferred burden will slow down development if commodity prices do not cooperate by diverting cash away from the drillbit.

For the royalty interest valuation I made some assumptions around individual well EURs and average wells per drilling unit. I created three cases: Low, High, and Upside with 18, 24, and 27 wells per section, respectively. You can see in the tables below how I got to those numbers. On the high side a prospective drilling unit might have up to 32 wells across several benches and on the low side 8 wells. The different inventory cases assume different ratios of the two within TPL's prospective acreage.

|                   | Wells per Section Estimate |      |         |
|-------------------|----------------------------|------|---------|
|                   | Low                        | High |         |
| Wells per Section | 8                          | 32   | Blended |
| Low               | 60%                        | 40%  | 18      |
| High              | 35%                        | 65%  | 24      |
| Upside            | 20%                        | 80%  | 27      |

For the Low Inventory case and High Inventory case I assumed an EUR of 150Mboe/1k lateral ft. In the Upside case I increased that to 200Mboe/1k lateral ft. In the table below you can see the math to get to an estimated EUR for TPL's royalty interests ranging from ~350MMBoe to almost 700MMBoe in the upside case.

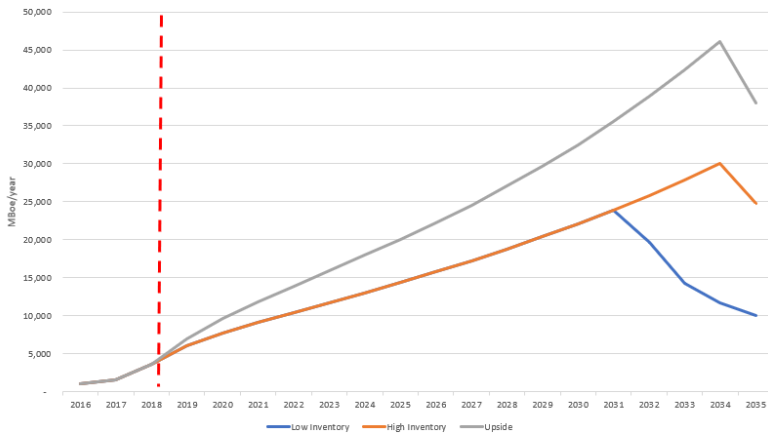
| Well Assumptions                   | Low Inventory | High Inventory | Upside     |
|------------------------------------|---------------|----------------|------------|
| Assumed EUR / 1k lateral ft (Mboe) | 150           | 150            | 200        |
| Lateral ft per section             | 5,000         | 5,000          | 5,000      |
| EUR per section per well (boe)     | 750,000       | 750,000        | 1,000,000  |
| avg wells per section              | 18            | 24             | 27         |
| EUR per section (boe)              | 13,500,000    | 18,000,000     | 27,000,000 |
| Acres per section                  | 640           | 640            | 640        |
| EUR per acre (boe)                 | 21,094        | 28,125         | 42,188     |
| Net Royalty Acres (100% NRI)       | 24,100        | 24,100         | 24,100     |
| Prospectivity Factor               | 68%           | 68%            | 68%        |
| Risked Net Royalty Acres           | 16,388        | 16,388         | 16,388     |
| Estimated TPL EUR (MMBoe)          | 346           | 461            | 691        |
| % Oil                              | 45%           | 45%            | 45%        |
| % Gas                              | 55%           | 55%            | 55%        |

Using the above total EURs and individual well EURs I forecasted estimates for TPL's oil and gas production from 2019 through 2058. I assumed 7.5% average production tax, 15% depletion shield, and a 21% corporate tax rate. I assumed a \$4/bbl discount to WTI and 50c/MMcf premium for gas to account for the liquids value. The table below lays out the after tax PV8 of the different inventory cases and sensitizes WTI price (\$2.75HHUB assumed for all).

#### After Tax PV8 of Royalty Interests

|                    |          | Wells per Unit |          |          |
|--------------------|----------|----------------|----------|----------|
|                    |          | 18             | 24       | 27       |
| WTI Price (\$/bbl) | \$ 50.00 | \$ 3,457       | \$ 4,137 | \$ 5,938 |
|                    | \$ 60.00 | \$ 3,957       | \$ 4,734 | \$ 6,794 |
|                    | \$ 70.00 | \$ 4,457       | \$ 5,332 | \$ 7,649 |

At \$60/bbl WTI the value of the royalty interests ranges from \$4.0bn to \$6.8bn depending on the inventory and EUR assumption. It is worth emphasizing that I am forecasting significant production growth here. This is the bulk of TPL's value, and if production growth starts to disappoint, that is a major red flag. The graph below lays out my modeled production growth (graph cuts off at 2035, but DCF runs through 2058).



#### Valuation

TPL's current TEV is ~\$6.0bn with no debt and \$93mm of cash. The table below shows the implied value of the royalty interest based on the after tax high and low cases for the surface rights and the water business.

#### Implied Value of Royalty Interests

|                      |      | Surface Acreage Value |          |
|----------------------|------|-----------------------|----------|
|                      |      | Low                   | High     |
| Water Business Value | Low  | \$ 4,941              | \$ 4,771 |
|                      | High | \$ 4,632              | \$ 4,462 |

If we assume the Low case value for both the surface acreage and the water business then the royalty interest value implied by the current TEV is ~\$4.94bn (top left quadrant above). Compared to the Low Inventory (18 wells per unit) \$60 WTI royalty value of \$4.0bn that implies ~17% downside to TPL. However, if we take the High case values for the surface acreage and water business (bottom right quadrant above), the implied royalty interest value is \$4.46bn. Using the the Upside Case (27 wells) there would be 39% upside to the current TEV. The two tables below show the upside/downside to TPL under the different inventory cases and WTI price assumptions. The first table assumes the Low case value for both the water business and the surface acreage, and the second table assumes the High case for both the water business and the surface acreage.

#### LOW Surface & Water Business Values - Implied Upside/Downside to TPL

|                    |          | Wells per Unit |      |     |
|--------------------|----------|----------------|------|-----|
|                    |          | 18             | 24   | 27  |
| WTI Price (\$/bbl) | \$ 50.00 | -25%           | -14% | 17% |
|                    | \$ 60.00 | -17%           | -3%  | 31% |
|                    | \$ 70.00 | -8%            | 7%   | 45% |

#### HIGH Surface & Water Business Values - Implied Upside/Downside to TPL

|                    |          | Wells per Unit |     |     |
|--------------------|----------|----------------|-----|-----|
|                    |          | 18             | 24  | 27  |
| WTI Price (\$/bbl) | \$ 50.00 | -17%           | -5% | 25% |
|                    | \$ 60.00 | -8%            | 5%  | 39% |
|                    | \$ 70.00 | 0%             | 15% | 54% |

You can argue about the right long term WTI price to use, but I am willing to bet on \$60 WTI, despite being a little higher than the current strip. At that price the risk/reward ratio for TPL appears to be favorable at the current stock price, especially given the tendency for the acreage in its neighborhood of the Delaware to get better over time.

I would highlight that the water business and surface acreage combined after tax valuations account for 17% (low cases) to 25% (high cases) of TPL's TEV. This is meaningful; however, the royalty interests account for the bulk of TPL's value

### Governance

There is a proxy fight going on right now between management and longtime holder Horizon Kinetics ("HK") who owns ~23% of the stock. I am rooting for HK, and their presentation is worth a read. Even if HK loses, they have made themselves heard, and the mgmt candidate has committed to a 3 year term (trustees normally serve for life) which is a major win for shareholders. The TPL CEO, Tyler Glover, is young and has no experience running a multi billion dollar business. I would prefer someone with more experience allocating capital at the helm. Furthermore, maximizing TPL's innate competitive advantages in the water business will take significant managerial skill. Fortunately, Tyler appears to be surrounding himself with experienced operators. Mismanagement is a real risk here given the lack of shareholder ability to effect change; however, the asset value is there.

### Risks

1. Oil and Gas activity / commodity prices / long term oil demand destruction
  - a. The majority of value here is driven by a forecasted 5x - 10x increase in oil and gas production over the next 10 to 15 years. The rock is great, but this is a bet on commodity prices and the global economy's continued reliance on oil.
  - b. The oil price risk is partially offset by latent long term upside in natural gas prices as there are significant natural gas reserves in this rock, and I am willing to bet we are at or near the floor for the domestic gas strip (famous last words).
2. Governance / management
  - a. We are forecasting significant growth in the water business. This will take skill to execute. So far so good, but this is a real risk despite the tailwinds.
  - b. The current proxy fight has been beneficial for shareholders no matter how it turns out, but at the end of the day, even if HK wins, a majority of the trustees will still be old guard, and shareholders have will still have little ability to effect change for the next 10+ years.
  - c. Significant cash flows will be generated over the next few decades. Current management has no history of allocating large \$ amounts. My hope is that they stick to the stated purpose of the Trust, which is to return capital to shareholders. But there is no guarantee and recent actions are not encouraging.

I do not hold a position with the issuer such as employment, directorship, or consultancy.

I and/or others I advise hold a material investment in the issuer's securities.

### Catalyst

Continued significant oil and gas production growth

Continued capital efficient growth of the water business

## Messages

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**Subject** Couple Questions  
**Entry** 05/09/2019 04:17 PM  
**Member** Light62

Thanks for the write-up. I think it's good - it lays out the case clearly and deals with the important questions fairly. Had a few questions though:

- 1) How much of the PV8 of the royalty interests comes after 2040 (as a somewhat random year) and how comfortable do you feel about underwriting oil demand 20+ years from now?
- 2) Why are you taking a depletion shield on the royalty interests?
- 3) One thing I've always struggled with with these on-the-come high-growth Permian names is what the contemplated growth means for aggregate production & thus the cost structure in the area (PXD's 1mm BoE/D target always struck me as very hard to fathom). As such: What do you think a 10-fold increase in TPL attributable production means for aggregate Permian production? What does that mean for Texas O&G cost structures and infrastructure needs? Do you think that's reasonable or does it mean that other basins or other countries will be more competitive before we hit the production levels needed to justify TPL?
- 4) Somewhat related to #3 but I've always been a bit concerned about capitalizing the easement income (non-water portion) at anything but a very low multiple. If this is largely upfront payments and we've seen a huge blowout of infrastructure construction in the Permian might we see these level off or even fall? I acknowledge that 10-14x doesn't contemplate much growth (especially in this market) but these payments probably can't be considered perpetuities the way some operating businesses can.

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**Subject** Royalty Interests  
**Entry** 05/09/2019 07:01 PM  
**Member** JL Gotrocks

For royalty interest NPV math, you didn't explicitly take into account what's been produced historically. Your highest productivity % acreage is also the most drilled up today, which makes the potential impact of this oversight much greater and therefore very difficult to have any confidence in your valuation. These are not virgin lands and cannot be valued as such. If you've done this work and for some reason didn't include it could you please share?

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**Subject** Re: Royalty Interests  
**Entry** 05/10/2019 12:19 AM  
**Member** JL Gotrocks

I've owned this company off and on for past five years. I don't own it today.

I suggest you take this post down as it includes a very large error and then once updated I'd imagine you end up wanting to short or being agnostic today, even at your \$60/bbl long term oil price assumption.

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**Subject** Re: Couple Questions  
**Entry** 05/10/2019 09:01 AM  
**Member** Plainview

Thanks Light62. Good questions.

- 1) 93% of the PV8 of the royalty interests is generated from 2019 -2040 (78% of estimated EUR produced by then). As you know there is a wide range of views on long term oil demand. My bet is that we will probably see continued global oil demand growth through at least 2035 with a meaningful plateau thereafter. As long as oil demand is growing, I think \$60 is a reasonable bet. The price required in a flat demand world 15 years from now is a question mark. However, if five years from now oil is still at \$60 and TPL's royalty production grows in line with my forecast and global demand growth still looks like it has a 10 year runway, I think this investment will have worked out nicely.
- 2) Royalty owners get to take a 15% depletion deduction. It does not run through TPL's GAAP income statement, but per the 2018 10-K "For Federal income tax purposes, however, deductions are made for depletion, computed on the statutory percentage basis of income received from royalties"
- 3) My High Inventory case forecast has TPL's royalty oil production increasing from 5kb/d in 2018 to peak in 2034 at 37kb/d. The avg NRI of 5.5% (taking prospective NRA's divided by prospective gross acres) would imply gross peak production from TPL's acreage of 667kb/d. PAA's fundamentals team sees 2023 Permian production of ~6.5mmb/d. Assuming that is the Permian peak production and it holds flat from there, that would imply ~10% "market share". That seems reasonable given the quality and quantity of acreage. My guess is Permian production peaks higher than that (but before 2034), but its a helpful statistic. My Upside case implies peak gross production of ~1mmb/d or ~15% "market share", which is more aggressive but not completely crazy. TPL's acreage in the Delaware is in the early innings compared to the more mature Midland basin (and of course the CBP) so its growth rate will be greater than the basin as a whole. The gross dry gas implied future "market shares" using this same methodology are a little higher (12 - 20%), but this is some of the gassiest acreage in the Permian so it will account for a disproportionate share of the gas growth. There will be significant pipeline infrastructure required to accommodate all of this growth. PAA, ET, EPD will benefit.
- 4) I agree with your concern regarding capitalizing easement income. Those multiples are outputs, not inputs. TPL has benefitted from a major infrastructure buildup recently, but I would argue that the easement income will continue for 10+ years given the number of gathering pipelines and well pads to be built for all of these future wells. Furthermore, a lot of the current easements will expire at that point which will generate additional income. Basically, your point is valid, but I think my figures are reasonable, and if I am off by 20% on the surface acreage value, it does not change the thesis.

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**Subject** Re: Royalty Interests  
**Entry** 05/10/2019 09:12 AM  
**Member** Plainview

I guess I know who gave this write up a 1, lol. Interestingly, if you add up all of TPL's royalty production from 2012 to Q119 you get 9.5MMBoe. That is 2.7% of my low inventory case estimated EUR (1.4% of the Upside Case). It is just not material. I guarantee that my forecast is off by a lot more than 2.7%, but it is not because I am ignoring the production to date. But you are right that I should have explicitly addressed this in the write up to avoid confusion.

Note that I chose 2012 as the starting point here because that is when the tight reservoirs that I tried to quantify in my analysis started to be developed with modern horizontal drilling and frac jobs. But if you go back a few more years, the answer does not change.

**Subject** Re: Re: Re: Royalty Interests  
**Entry** 05/10/2019 11:46 AM  
**Member** JL Gotrocks

Good example of how using averages can get a guy in trouble I think. Look at the counties you're giving the most value to - Culberson, Reeves and Loving. 2018-2012 gross wells 140 Culberson, 161 Reeves and 221 Loving on 1/16. ~90% prospectivity ranking on Reeves doesn't make any sense to me given approximately half of this acreage is Alpine High gas lands which had been getting deferred with Waha price weakness. Adjusted for prospectivity implies 2018-2012 drilled up to two wells per section already, which is **7-11% of your 18-27 wells per section**. But you can't stop here, because what happens in every field is **the best lands get drilled first**, so the remaining lands are sure to be lower density spacing than what's already been drilled (18-27 for remaining likely much too high), lower prospectivity rock (economics are worse than what's already been drilled), often takes from parent well assumed reserves etc. This is often very material, like that ~10% miss on your math could easily be 20% lower.

**Subject** Re: Re: Re: Royalty Interests  
**Entry** 05/10/2019 04:12 PM  
**Member** Plainview

I appreciate the pushback JL Gotrocks, but I do not think any of your points are valid.

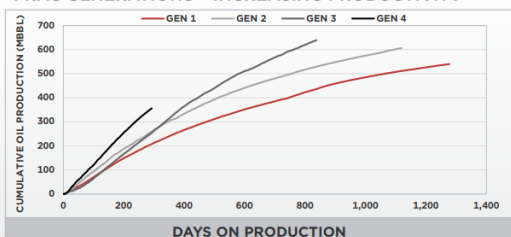
1. Looking at the map of TPL's Reeves NPRI acreage (not surface acreage), I estimate at least 70% is core oily acreage in the northern and eastern portion of the county. The other 30% is Alpine High. All of this acreage is prospective. I gave 90% credit instead of 95% to account for the fact that the Alpine high wells will not generate as much revenue as the oily wells (at the current price deck). Also, my 45% oil type curve was set to take into account the fact that some areas, such as the Alpine High, are gassier than others. Most of Loving county and Reeves county wells will be >50% oil.

2. You are misapplying your well count figures to reserve calculations. Let's take your Loving County example. 71 of those 221 Loving County wells were drilled in 2018. Lets assume they were completed on average at 6/30/18. That means that as of the balance sheet date 3/31/19, they had been producing for nine months. At nine months, I estimate a well has produced around 24% of its EUR. So those 71 wells really only decrease my effective location count (in terms of reserves) by 17 (.24\*71). If I apply the appropriate cumulative EUR produced % to each year's well vintage (37% for 2017, 45% for 2016 etc) then the 221 well locations drilled in Loving since 2012 that you mentioned gets decreased to 85. That would be ~1 well per section that has been "drilled up" already which is 4% to 6% of my 18-27 wells per section. Performing the same exercise to the wells drilled in Culberson and Reeves, I get 0.5 wells and 0.3 wells per section respectively. Thats 1% to 2% of TPL's inventory in those counties. The weighted average of all three counties is 0.56 wells "drilled up" per section". Thats 2% to 3% of my estimated TPL inventory. These figures are exactly in line with my back of the envelope math in comment 5.

3. Delaware basin wells have continued to get better every single year since activity accelerated in 2016/2017. IP/1k ft and cumulative production/1k ft have increased dramatically as optimal completions have become better understood. Look XEC's chart below from its most recent presentation showing the cumulative oil production improvement due to improved frac designs. It is material. Are there parent/child well issues in some places? Absolutely. But the wells overall are improving, and those claiming otherwise have been proven wrong over time. Check out what CLR is doing in the Bakken. That play is way further along than the Delaware in its development. A large % of the good rock has been drilled. So now CLR is stepping out into "Tier 2" areas with the most up to date completion techniques and drilling great wells. This will happen in the Delaware as well.

## Culberson: Resilient Long Later

FRAC GENERATIONS - INCREASING PRODUCTIVITY



**Subject** Re: Re: Re: Re: Royalty Interests  
**Entry** 05/11/2019 12:45 AM  
**Member** JL Gotrocks



1. Zoom in a count up the squares. What your saying is plain wrong.
2. So a well drilled today, isn't worth anything g more tomorrow? How does pdp production tomorrow account into your calculations. This is a very basic concept even 1 month reserve analysts out of school understands. OOIP and recoverable per section is a concept worth exploring...
3. Yeah we all get more sand and longer laterals have generated greater recoveries. Reality of drilling best lands first is that ... if you don't get it by now you have no hope ...

your numbers are at a minimum arithmetically 10% incorrect. Conservatively 20% off and more likely 30%. This is not typical variation. This is just being wrong about your miss understanding of basic concepts.

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| <b>Subject</b> | Re: Re: Re: Royalty Interests |
| <b>Entry</b>   | 05/11/2019 12:56 AM           |
| <b>Member</b>  | JL Gotrocks                   |

How it works is that there's a certain amount of oil in place per section. If you have a well in that section that's producing today, what's left for the future wells has to take into account what that past well will produce in the future. It's so ridiculous to say this, but just taking into account what's produced historically ignores that the wells producing today will continue producing tomorrow, and that production means less recoveries for tomorrow's drilling. It's simple. Yet you ignore this simple but very true and basic idea. Your math at the most conservative level is ~10% wrong, and more likely be be arithmetically and just logically, not even subjective assumption based, at least 20% just plain wrong.

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| <b>Subject</b> | Re: Re: Re: Royalty Interests |
| <b>Entry</b>   | 05/11/2019 07:00 PM           |
| <b>Member</b>  | Plainview                     |

JL Gotrocks - I apologize if I am being dense here, but from my read of your comments, I must not have explained my methodology clearly enough. So I will try to do so again, one more time. Using assumed prospectivity %'s and an assumed average EUR/1k lateral ft, I estimated a total EUR for TPL's royalty interests. I believe my assumptions are reasonable to conservative, but it is possible for people to have different opinions on those assumptions. I then created a PDP forecast from Q119 using Arps decline curve analysis. On top of that PDP forecast, I layered a drilling wedge of new wells completed over the next 12 to 15 years. I forecasted sufficient well completions such that sum of the PDP EUR and the new well drilling wedge EUR was equal to my estimated total EUR for TPL's NRI. In comment 2, you pointed out that I was not accounting for historical production in my analysis. And in comment 3, you declared my analysis so wrong that the write up should be taken down. In comment 5 and in comment 7, I laid out two different methodologies for calculating total historical production. Both methodologies indicate that through Q119, 2% to 3% of my estimated EUR has been produced already. My view is that this 2% to 3% is immaterial to the analysis. To demonstrate why 2% to 3% is immaterial to this analysis, take my type curve assumption. I believe that while retaining the spirit of conservatism and without compromising the analytical rigor of the forecast, I could have easily chosen 155 Mboe / 1k lateral ft instead of 150 Mboe / 1k lateral ft, a ~3% difference. If I had assumed a 155 MBoe / 1k lateral ft type curve and then deducted ~3% of the total calculated EUR to account for historical production, my outputs and conclusions would exactly be the same, but we would not be having this discussion.

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| <b>Subject</b> | Re: Re: Re: Re: Royalty Interests |
| <b>Entry</b>   | 05/12/2019 08:51 PM               |
| <b>Member</b>  | JL Gotrocks                       |

Plainview - thank you for laying this out clearly as I believe you are trying to the best of your abilities to put a reasonable piece of analysis forward. Maybe I have not communicated what you're missing in an effective way in prior posts, so please let me try once more. This will be my last post on this thread.

An undeveloped play analysis that starts with EUR per lateral foot, ignores what's been developed on the lands in the past and overlays a per well spacing assumption on the EUR per lateral foot assumption is subject to very significant undue error potential. The common place to start is with OOIP and then to take this perspective on a risk analysis section-by-section. There is only so much oil in a section, only so much prospective per zone, and only so much of that oil in place that is reasonable to be assumed to be recovered on a certain spacing scheme. Offsetting wells, logs, historical production, depth, thickness and more all go into it.

One of the easiest and most straightforward places to start for any public market investor trying to determine what is reasonable for future drilling inventory, is to take what's already been produced out of your future recoverable resource estimate from new well drilling as it's already been produced (which you admit is something that should be counted however it is just 3% so not material if missed), or will be over the remaining life of the well (the part of it you ignore). If you only take what's been produced to today out of future recoverable from new well drilling, you miss the remaining reserves to be produced from existing wells. This is important to include and why there's so much fuss about spacing and child-parent relationship. Your math implies a 3% error on your undeveloped valuation because you only took into account historical production. Taking into account the reserves that are not yet produced today but can be expected to in the future from existing wells, the difference is much larger than 3%.

Now take this back to the grounding idea that there is only so much oil in place and reasonable assumed to be recoverable per section and that the best acreage gets developed first, and it has large implications from a remaining play development perspective. Take for example Spraberry trend development on Texas Pacific's Midland acreage, which had 127 wells drilled 2012-2018 and on your 95% risk factor implies ~7 wells per section drilled today (half of which in 2018, and much likely in H2 so historical production from these wells is very likely to be less than future recoverable of these producing reserves). What that means using your assumptions, is that current wells into the Midland Spraberry are expected to recover 5.25MM to 7.00MM per section, which on your 13.5MM to 27.0MM total is 26-35% of the total EUR per section you assume to be recovered. Your analysis as it stands completely ignores this. Ignoring this also has further implications for the analysis in that your 24 wells per section of future drilling inventory becomes much riskier than you had thought (you're implicitly assuming over 30 wells per section get developed on your mid-case). Because the Spraberry has already been drilled up, but you didn't take this into account, a whole ton of value you're giving on an 8% discount rate is associated with far less prospective and riskier zones, on tighter spacing, that the operator has already passed up on and which your type-curve is far less certain to match.

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**Subject** Re: Re: Re: Re: Re: Re: Royalty Interests  
**Entry** 05/13/2019 08:00 AM  
**Member** Plainview

This is why I would have been a terrible banker. Apparently, I am a truly awful communicator. I am not sure why I have been unable to disabuse you of the notion that I am somehow ignoring future production from existing wells. I like to think of myself as an optimist though. So one more time, with bold font for emphasis. **My forecast takes into account future production from existing wells.** I forecast the PDP production curve and layer on top additional new wells such that only my calculated total EUR ever gets produced from the reservoir, including the PDP volumes. The only volume I am not taking into account is the production to date, which I have shown to be de minimis at 2 to 3% of the calculated EUR. I do not know how else to explain it. Maybe if I had the aid of a powerpoint slide deck...

Best of luck to you going forward, JL Gotrocks.

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**Subject** Re: Re: Re: Re: Re: Re: Royalty Interests  
**Entry** 05/13/2019 09:29 AM  
**Member** JL Gotrocks

I too am an optimist so will try to show once more in one sentence. Taking an arbitrary eur per lateral assumption and multiplying it by an arbitrary number of wells per section on this land base is pure fantasy.

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**Subject** Re: Re: Re: Re: Re: Re: Royalty Interests  
**Entry** 05/13/2019 09:53 AM  
**Member** Plainview

Progress! This is in fact a valid critique of the write up, albeit a different critique than the one you have been pushing to this point. That said, I strongly disagree with your assessment that my EUR and well assumptions are arbitrary. And I do believe that the methodology I have presented is actually very helpful in framing the distribution of possible outcomes.

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**Subject** Re: Re: Re: Re: Re: Re: Royalty Interests  
**Entry** 05/13/2019 10:10 AM  
**Member** JL Gotrocks

This is great! ok so now that we're close to being on the same page, what isn't clear about how a section that's already drilled up doesn't have as many remaining locations as an undeveloped section of similar quality (that's one way in which my last comment and all the comments prior are tied)?

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**Subject** Re: Re: Re: Re: Re: Re: Royalty Interests  
**Entry** 05/13/2019 11:04 AM  
**Member** Plainview

I am going to try a different format here.

Total Net Locations = Net PDP Locations + Net Undeveloped Locations

Total Net EUR = Total Net Locations \* EUR per Location

Total Forecasted Royalty Revenue = Commodity Price \* Total Net EUR

The only piece I am not explicitly taking into account is the production from the PDP locations prior to the model start date which would decrease the EUR per PDP location. However, I have shown that amount to be de minimis (2% to 3% of Total EUR).

To reiterate, I am not modeling 18 - 27 undeveloped locations. I am modeling 18 - 27 total locations.

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|----------------|-----------------------------------------------------------|
| <b>Subject</b> | Re: Re: Re: Re: Re: Re: Re: Re: Re: Re: Royalty Interests |
| <b>Entry</b>   | 05/13/2019 11:26 AM                                       |
| <b>Member</b>  | Light62                                                   |

I may regret doing this but I've been reading this conversation because JL's comment struck me as important & I was surprised Plainview's model returned the values for the royalties that it did and, assuming Plainview's response doesn't put it to bed, would explain the confusion here as:

JL believes that Plainview is:

1) Assuming no wells have been drilled on any of the TPL acreage in his forward development model (i.e. he gets 18-27 wells on every section in the future)

AND

2) Also assuming that currently producing wells have value to TPL.

His concern is there is double counting to the extent that any wells in #2 should be subtracted from #1. This kind of double counting is very substantial and may represent more than teens+ % of royalty value.

Plainview says his model:

1) Assumes no wells have been drilled on the TPL acreage (i.e. he gets the full 18-27 wells per section in his development model)

BUT

2) Does NOT include future production from currently producing wells in his valuation

Doing it this way does lead to some double counting but only to the extent that the portion of oil already produced from already drilled wells is effectively counted twice (and has already been divided out). The tail of production from these wells is only counted once. This double counting is not very significant and represents a low to mid single digit % of total recoverable resource.

My one question for Plainview is - if you aren't assuming any future value from currently producing wells how does your production forecast chart run so smoothly from current production levels? Do you just back into what kind of drilling over the next quarter/year would be required to bring TPL's production in-line (along with some healthy growth) with historic production?

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| <b>Subject</b> | Re: Re: Re: Re: Re: Re: Re: Re: Re: Re: Royalty Interests |
| <b>Entry</b>   | 05/13/2019 12:00 PM                                       |
| <b>Member</b>  | Plainview                                                 |

Light62 - I greatly appreciate you wading in here, but unfortunately, you have misstated what I have done. Since two different people have not followed my assumptions, I have clearly been remiss in explaining my model. I am not sure if you saw my last comment before posting your comment, so I will rehash it in part.

Total locations = PDP locations + undeveloped locations

Total EUR = EUR per location \* Total locations

After calculating an estimated Total EUR for the royalty interests, I created a PDP forecast for the existing wells from Q119 forward using Arps decline curve analysis to get a PDP EUR.

On top of that PDP forecast I layered a drilling wedge of new wells completed over the next 12 to 15 years. In the drilling wedge I modeled sufficient well completions such that the sum of the PDP EUR and the new drilling wedge EUR was equal to my estimated Total EUR for TPL's acreage. So the Total EUR produced in my model = the Total EUR calculated above. Part of that EUR comes from a PDP forecast and part of it comes from the drilling wedge.

The reserves I am not explicitly accounting for in my model are the reserves produced to date by the PDP wells. This means that I am effectively double counting the 9.5MMboe that has been produced from the PDP wells as of Q119. If I had explicitly accounted for the 9.5MMboe in my model, it would have cut off ~13 net completions (100% NRI to TPL; 9.5MMboe/0.75MMboe per location) from the back end of the drilling wedge in the year 2031 or 2034 depending on the case and decreased the value of the royalty interests by ~1%.

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| <b>Subject</b> | Re: Re: Re: Re: Re: Re: Re: Re: Re: Re: Re: Re: Re: Royalty Interests |
| <b>Entry</b>   | 05/13/2019 08:14 PM                                                   |
| <b>Member</b>  | Plainview                                                             |

I wanted to add a little additional color to my previous comment that might help clear up some of the confusion. The locations per section assumption I made was to calculate a Total EUR to be recovered from the reservoir. That location count does not inform how many wells are drilled in the forecast. The number of wells drilled in the forecast was set so that my forecasted PDP EUR + the drilling wedge EUR was equal to my estimated Total EUR. While we do know how many wells have been drilled on TPL's acreage, and we can estimate their production to date, we do not know what the actual total completed lateral feet of those wells was on TPL's acreage. That is why I have not tried to match my modeled location count to my assumed location count. Instead, I simply drilled the number of 150Mboe/1k lateral ft type curve wells necessary for my PDP forecast EUR + drilling wedge EUR to equal my calculated EUR. As we have discussed, I did not deduct the 9.5MMboe produced to date from my calculated Total EUR and therefore my model "double counts" those reserves by including them in the forecast despite having already been produced. As I explained in my previous comment, the way I would fix this overstatement under my methodology would be to cut off ~13 net completions from the back end of my drilling wedge forecast which would decrease forecasted EUR by the 9.5MMboe already produced from the existing wells. Doing so decreases my royalty interest valuation by ~1%.

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| <b>Subject</b> | Re: Re: Re: Re: Re: Re: Re: Re: Re: Re: Re: Re: Re: Royalty Interests |
| <b>Entry</b>   | 05/14/2019 09:30 AM                                                   |
| <b>Member</b>  | AdmiralJames609                                                       |

I was about to comment on the earlier thread that this "double counting" allegation ultimately pertains to a negligible portion of total NAV. Thank you for explicitly clarifying it here... in fact a <\$0.30 shift in your BOE price assumption would have about the same NAV impact as removing the resource that has already been produced from your calculation.

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| <b>Subject</b> | Re: Re: Re: Re: Re: Re: Re: Re: Re: Re: Re: Re: Re: Royalty Interests |
| <b>Entry</b>   | 05/15/2019 01:23 AM                                                   |
| <b>Member</b>  | JL Gotrocks                                                           |

Judging by past responses if I attempt to keep going on the same path the core of what I'm saying will be ignored and deflected for what I can assume to be an attempt to save face, but so be it. I appreciate you engaging with me here. I only bring all this up because I believe with a very high degree of certainty this work is fundamentally flawed resulting in difference far greater than the sub 5% claimed, I hold VIC to a high standard, and believe engaging in a comment section is far better for quality of analysis on this forum than being critical in anonymous tags (which is the norm, very common and is far more rude than either of us have been here).

Perhaps I'll try to approach my original concerns another way, this time by asking just a few simple questions.

A mid-case of 24 wells per section, across two different basins, dozens of counties, many operators with varying development plans and what I would think is well over fifteen zones developed changing drastically across the land base in prospectivity, deliverability, oil mix and more.

- How did you approach building your weighted average type-curve? Can you be more specific on type curve assumptions (e.g. decline, oil mix through time, first year production) and how you built it up?

- How did you approach risking wells per section (8 to 32) and assumed prospectively per county (0 to 95)?

- How does the two above tie together in your mid-case (think you call it high?) development plan and assumptions generally?

Thanks.

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| <b>Subject</b> | Re: Re: Re: Re: Re: Re: Re: Re: Re: Re: Re: Re: Re: Royalty Interests |
| <b>Entry</b>   | 05/15/2019 06:13 PM                                                   |
| <b>Member</b>  | Plainview                                                             |

The core of what you have said to this point is that there is an internal inconsistency in my methodology that causes my outputs to be "arithmetically 10% incorrect. Conservatively 20% off and more likely 30%". In fact, you stated that the write up is so wrong that it should be taken down due to this alleged error. The way you tried to explain my alleged error made it clear to me that I had failed to explain my methodology adequately. I am not sure if I have succeeded yet in my attempts to explain my methodology more clearly for you, although AdmiralJames609's comment makes me believe that there is hope for me yet.

To be clear, in no way have I or will I attempt to save face. If in the course of our discussion you point out an error in my methodology, my logic, or my assumptions, I would truly be ecstatic as I would much rather find out on this message board than by losing money on the investment. The point of this entire exercise is to make money, not to prove myself right. Based on the questions you have asked above, it appears that we can now move on to the substance of the thesis, and I sincerely hope that you can point out errors in my work.

Let me first respond to your "dozens of counties" claim because I believe it is important to understand the scope of the bet. Reeves, Loving, and Culberson account for 83% of my estimated royalty interest value. Adding Midland, Upton, Glasscock, Pecos and Howard to those three gets us to 97%. This is a big bet on three counties, and eight total counties account for the overwhelming majority of the royalty interest valuation. That is still a whole lot to get your arms around, but it is very different from "dozens of counties".

To state the obvious, wells vary widely across TPL's acreage. On the same acreage across multiple zones there are different oil cuts and EURs. Lets start with my oil cut assumption of 45% first. I started with TPL's historical production, which in 2018 was 47% oil. Since I am modeling such a large increase in future production, that is not necessarily a good estimate. APC's 10-K does not break out reserves by area, but their 2018 Delaware production was 60% oil. I then looked at the YE18 reserves report for 10 pure play public Permian operators. For the four Delaware players (MTDR, JAG, CDEV, HK), proved reserve oil cut ranges from 55% to 77%. The two Midland players (PXD, and LPI) were 54% and 26%. The four Midland/Delaware combo players (PE, CPE, FANG, CXO) ranged from 56% to 76% oil cuts. CVX's current total Permian production is ~52% oil. APA's Alpine High locations are only 15% oil. The overlap here with TPL is not perfect, but the average %'s are indicative. All of this does not give us a great answer, but given the datapoints, 45% oil cut on average feels conservative.

I performed a similar exercise for EUR per 1k lateral feet. A lot of companies are hesitant to provide EUR estimates because they are difficult to quantify with precision and are frankly less meaningful to the well economics than the first few years of cumulative production. However, to model the NAV, EUR estimates must be made using whatever information is available (company statements, research reports, etc). The oily type curves (>50% oil) in the Delaware range from 150MMboe/1k lateral feet to >250MMboe/1k lateral feet, with XEC on the high end of that range (Culberson, Upper Wolfcamp; ~50% oil). Taking Alpine High as the gassy type curve example, their "typical well" rich gas average type curve EUR is ~270 Mboe / 1k lateral feet (85% gas). The upper range Alpine High well is double that. PXD's type curve in the Midland Basin is ~140MMboe / 1k lateral feet. Based on the above figures, I believe that the 150MMboe per 1k lateral feet EUR and 45% oil cut is conservative and 200MMboe per 1k lateral feet is reasonably aggressive.

I would note that the oil EURs per 1k ft above range from ~40Mbo to >125 Mbo with the majority lying in the 75 to 100 Mbo range. Pick some reasonable % splits, and I think you are >85 Mbo per 1k lateral feet for the weighted average. I am modeling 67.5 to 90 Mbo per 1k lateral feet. I am not modeling an increasing GOR over time. I believe this is a conservative assumption as doing so would either necessitate an increase in gas EUR or bring forward in time higher revenue oil production.

For my TC production profile, I assume a 75% initial decline and a 1.1 B factor. Cumulative production %'s by year are Year 1 - 24%, Year 2 - 37%, Year 3 - 45%, Year 4 - 51%, Year 5 - 56%, Year 6 - 60%. I cut off the type curve at 30 years.

One other relevant note on type curve. After coming up with my first cut based on the research summarized above, I spot checked it another way. I forecasted a PDP decline curve starting from YE 2016. I calculated the net wells drilled in 2017 and 2018 on TPL's acreage (100% NRI), assuming 5k completed lateral feet per well (11 in 2017 and 18 in 2018). I then started a 150MMbo per 1k ft drilling wedge on 1/1/2017 and layered on 11 wells in 2017 and 18 wells in 2018 to make sure that the 2017 and 2018 production profiles generated in the model matched what actually occurred in reality. The assumptions outlined above created a very close fit assuring that my 150 MMboe per 1k ft type curve is more or less in line with the average net well completed in 2017 or 2018.

Inventory Assumptions: APC has 33 identified locations per section on their Loving county acreage and recently successfully tested 20 well spacing in the Wolfcamp A alone. XEC recently successfully tested 14 well spacing in the Lower Wolfcamp in Culberson County and is running 8 to 12 well spacing tests in the Upper Wolfcamp this year. That is 22 to 26 wells in the Wolfcamp alone with upside from the 2<sup>nd</sup> and 3<sup>rd</sup> Bone Spring and Avalon formations. CDEV completed good wells in 5 zones in 2018 (some more delineated than others) in eastern Reeves County. FANG assumes 20 wells per section in premium zones in the Delaware and 28 wells per section in the Midland. Their 3 comparable peers (CPE, QEP, and PE) all assume >30 wells per section in the Midland and 24 to 29 wells per section in the Delaware. The % splits I laid out are simplistic, but I think that an average range of 18 to 27 locations per section provides a reasonable distribution of outcomes. As far as prospectivity, I looked at the map, I counted squares, and I made judgments based on wells results, heat maps, and commentary from all of the operators mentioned above.

For activity assumptions, I started with 2018 net well completions and escalated it at 7% per year starting in 2019. In the Low Case, activity is cut off at 12/31/2031 as that is the date at which sufficient wells have been drilled to produce my calculated EUR (including the PDP production). In the High Case, I assume 7% escalation per year and activity is cut off at 12/31/2034. In the Upside case I escalate activity at 8% per year and cut it off at 12/31/2034. In my opinion, activity level will be the single most important factor in determining the value and ultimate success of this investment. I believe that escalating 2018 activity level by 7 to 8% per year is a reasonable assumption given the rapid growth in production and cash flows generated by TPL's acreage. I would also point out that 2018/2017 activity growth was 67%.

**Entry** 05/17/2019 12:08 PM  
**Member** snarfy

The rock is great. Also got the surface rights. They are positioned in the bulls eye. The assets are great but let's set that aside for a minute. Basically you are paying 30x for a business where the people in charge own hardly any stock and a lot of potentially ugly self dealing has come to light. They have also given themselves big pay raises. Their interests seem squarely aligned with looting this treasure chest. I have heard from multiple sources they are almost impossible to deal with and/or incompetent. Their threats to Oliver are stunning. The trustees behave as if they have something awful to hide. The cockroach theory seems to apply here. How do you get comfortable this doesn't end badly for new investors paying current market prices?

**Subject** Re: Re: Re: Re: Re: Re: Re: Re: Re: Re: Re: Re: Re: Re: Re: Royalty Interests  
**Entry** 05/17/2019 08:21 PM  
**Member** JL Gotrocks

thanks for this. this is what i thought it was. you used assumptions from a few investor presentations and 10-Ks of operators in the general same proximity where you're giving tpl it's value. there wasn't actually a section by section build up, look back or close look of any kind on the land base itself.

**Subject** Re: Governance  
**Entry** 05/18/2019 08:58 AM  
**Member** Plainview

Governance is a major risk to the thesis. I think that you can make a reasonable argument that the trustees' and management's actions over the past couple of years, taken together as a whole, have been weak. I do not think their past actions indicate anything close to criminal. I do not think their past actions indicate that they have or will willfully disregard their fiduciary duties. The worst behavior seems to have been perpetrated by the former CEO, who has left the company. That said, no matter the outcome of the proxy contest, I absolutely believe that their actions going forward require very close scrutiny.

CEO and CFO compensation increased from ~\$700k each in 2017 to ~\$2.3mm each in 2018. I think the trustees could argue with a straight face that the establishment and buildout of the water business justified large bonuses, although I would certainly take issue with the size of the raise. But worse than that, the trustees have not even tried to put a compensation structure into place. Cash bonuses are paid based on the "overall impression" of performance. If compensation increases dramatically again in 2019, without reasonable justification, the thesis will have to be revisited. In fact, if a reasonable compensation structure (with explicit goals and target bonuses comparable to a peer group) is not put into place, or at the very least discussed, shortly after the resolution of the proxy contest, the thesis will have to be revisited.

I have also heard that TPL management is very difficult to deal with / incompetent, although the comments I have heard came from competitors in the water business, not from customers of the water business. I can understand why their competitors would not speak highly of a team that was gift wrapped such an enormous competitive advantage in chasing customers. Also those same competitors have to pay TPL for gathering line ROW, and I know TPL makes their lives very difficult.

The proxy battle has been ugly. Most are. People do not like to be told that they are idiots. But the trustees have responded poorly, and they look bad for it. I think that a three year term for General Cook would not be a bad outcome for shareholders. He has no relevant oil and gas experience, but he does have a history in public company corporate governance, which is what is most needed at this moment. I do not think anybody would say that General Cook is dishonest or that he is likely to neglect his fiduciary duty as he sees it. If General Cook is elected and does not appear to be asserting himself early and often on corporate governance issues (as promised), that is another major red flag.

The good news, and how I ultimately got sufficiently comfortable to make an investment, is that most of the future cash flow generated by the assets requires little to no action on the part of the trustees or management. The most important question to answer is will they distribute those cash flows to shareholders in the form of dividends or buybacks. The Trust indenture is clear that the trust exists for the sole purpose of liquidating its assets and distributing the proceeds to shareholders. The Trust indenture also makes clear that management has the ability to make investments to maximize the value of the Trust assets. Over the past 12 to 18 months, management has been focusing more on the latter. In some ways this is good. Maximizing the value of the water business absolutely requires capital outlays. So far they seem to be earning a very good return on those capital outlays. However, I strongly take issue with the fact that they reinvested meaningful asset sale proceeds in both royalty interests and land. More of that capital should have been distributed to shareholders, and I see no reason whatsoever for management to be investing in more royalty interests.

The stated purpose of the Trust should set a high bar for retaining and investing capital. If significant amounts of capital continue to be reinvested into royalty interests, or if the incremental returns on capital in the water business begin to fall, the thesis will be revisited immediately. In my opinion, this is going to be the most important signal as to whether governance is going to sink the investment.

It is a fine line. I would prefer that the current CEO and Trustees were not the ones entrusted to walk it. My thesis assumes that as cash flows continue to climb rapidly, management will return the vast majority of it to shareholders. I plan to continue to verify that assumption with every new data point I can gather.

**Subject** Re: Re: Re: Re: Re: Re: Re: Re: Re: Re: Re: Re: Re: Re: Re: Royalty Interests  
**Entry** 11/18/2019 02:27 PM  
**Member** JL Gotrocks

If you look at Texas Pacific's latest investor presentation (a lot more detail now), it's worth taking a look at the unrisks EUR by area from slide 16 (on RS's type curves, which may or may not capture child / parent issues...) and risking / filtering it for expected before tax IRR of over 38% at flat \$55/bbl (\$59 WTI on the diff assumption in the write up) and \$2.75/mcf pricing (slide 14). Think that's a reasonable filter since it's next to impossible to argue a BT IRR below this level generating any real corporate FCF or earnings or return after G&A, land, interest, options, seismic, facilities, failures etc. On Texas Pacific's own sunny day EUR estimate with their own 38% BT IRR filter at a price deck likely higher than what most use (anyone basing oil stock decisions on over \$60 WTI assumptions?), their net recoverable EUR is approximately inline with the "Low Inventory" case. Take the low inventory royalty value on AT present value at this 8% discount rate (think assumes a capital pace driving a 14% production CAGR for next 20 years on it...) with the midpoint water and surface value, and the stock is trading right on top of that today.

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|----------------|-----------------------------------------------------------------------------------|
| <b>Subject</b> | Re: Re: Re: Re: Re: Re: Re: Re: Re: Re: Re: Re: Re: Re: Re: Re: Royalty Interests |
| <b>Entry</b>   | 11/18/2019 07:26 PM                                                               |
| <b>Member</b>  | Plainview                                                                         |

JL Gotrocks - I agree with your general conclusion. The new investor deck is enormously helpful relative to TPL's prior disclosure efforts. The stock may perform well from here, but with this new information in hand, I cannot conclude that there is a compelling discount to NAV. Furthermore, as the Permian rig count continues to drop week after week, near to medium term production growth on TPL's acreage may disappoint. To stay long now, you would have to believe that as the Permian rig count drops, the remaining rigs will move disproportionately to TPL's acreage. To some degree, this migration will happen due to the high quality acreage and blue chip operators (Oxy fiasco notwithstanding). However, the lower the rig count goes, the longer that putt becomes. And there are cheaper ways to be long oil and/or gas, if that is the bet you want to make. Time to move on.

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|----------------|----------------------------|
| <b>Subject</b> | Author Exit Recommendation |
| <b>Entry</b>   | 11/18/2019 07:27 PM        |
| <b>Member</b>  | Plainview                  |

The author has recommended exiting the position

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|----------------|-------------------------------------|
| <b>Subject</b> | RE YOUR UINITIAL ANALYSIS VALUATION |
| <b>Entry</b>   | 12/06/2020 03:59 PM                 |
| <b>Member</b>  | dadande929                          |

WHAT WAS YOUR ALL IN TPL TOTAL VALUE OF THE TRUST IN \$BILLIONS AND PER SHARE?  
ASSUME P VALUE OF \$50

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|----------------|---------------------|
| <b>Subject</b> | JL GOTROCKS         |
| <b>Entry</b>   | 03/17/2021 08:05 PM |
| <b>Member</b>  | ele2996             |

Are you short in size with TPL at a record high? I certainly hope so.

Plainview, you nailed it. Sorry that you got swayed by a bully. It's always tough to take a position on a weird company.

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|----------------|---------------------|
| <b>Subject</b> | Re: JL GOTROCKS     |
| <b>Entry</b>   | 03/18/2021 07:22 AM |
| <b>Member</b>  | wjt                 |

I don't know much about TPL or this space, but besides higher oil prices and the governance reform, could someone share their thoughts on the recent stock price move?

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| <b>Subject</b> | Re: Re: JL GOTROCKS |
| <b>Entry</b>   | 03/18/2021 10:23 AM |
| <b>Member</b>  | nola18              |

Just my 2c... but the disconnect here is pretty absurd.

As is highlighted in this writeup, and others bullish on the name (i.e. Horizon Kinetics), the O&G royalty assets account for 75%+ of the value of TPL. VNOM is almost a direct comp to TPL. Each have ~100K net mineral acres (~24k net royalty acres). VNOM has substantially better acreage than TPL, and in fact, 4.5K of the 24k net royalty acres for TPL are central basin platform not Midland/Delaware. TPL is slightly less developed than VNOM, but it's not a huge delta. Right now TPL is trading at 4x VNOM. Makes no sense. Could be people focused on the bad hedge book for VNOM for the next 12 months, but that's a tiny fraction of the long term value here.

And now for a shameless BSM plug... In addition to 34k net mineral acres in the Midland Delaware, and ~200K net mineral acres in central basin platform... BSM has 260k net mineral acres in the Haynesville that generate about half as much cash flow as an equivalent acre in the Midland/Delaware. AND you have ~7m net mineral acres in 60+ other onshore basins around the country...Right now BSM is valued at 1/6th TPL.

Best,

Nola

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|----------------|---------------------|
| <b>Subject</b> | Valuation           |
| <b>Entry</b>   | 11/08/2022 09:36 AM |
| <b>Member</b>  | kgalt               |

this is a fabulous business, but are the multiples out of whack at this point?

30x sales

47x earnings

58x free cash flow

57x ebitda